ITE4052 Computer Vision (Fall 2026)

Assignment 2

- Proposal Report Deadline: November 15th 11:59 PM.
- Intermediate Report Deadline: November 29th 11:59 PM.
- Final Report Deadline: December 18th 11:59 PM.
- Submit a report summarizing the results (pdf) on the LMS — for the final submission, also include all code as a .zip file.
- You can use either English or Korean for the report.
- Late submission will get the half score.
- AI tools may be used only for assistance with understanding papers and with coding.
- Reports must be written entirely in your own words, based on your own understanding of the baseline and your idea.
- Low-quality reports (e.g., logical errors, inconsistent/inappropriate expressions, or lack of genuine understanding) will be treated as a violation and may result in a significant penalty.

Description of the Project

Project Overview

In this assignment, you will carry out **a small research-style project** on a computer vision topic of your choice. You will (1) choose a topic, (2) find a relevant paper (accepted at CVPR, ICCV, ECCV, AAAI, NeurIPS, ICML, ICLR, EMNLP, or ACM MM — the paper does not need to be from 2025–2026; papers from 2021–2024 at these venues are equally acceptable) that addresses the chosen topic and use it as your baseline, (3) select one of the benchmark datasets used in the baseline and reproduce the baseline's results on that dataset (a sampled/reduced version is fine if the full setting is too heavy), (4) analyze the limitations you observe from the reproduced baseline, and (5) propose and test your approach to address one of those limitations. **Your final method is required to outperform your reproduced baseline.** You can refer to follow-up works of the baseline as long as they are cited, but we will give additional credit when the improvement comes from your own creative method.

The project is submitted through three cumulative reports — Proposal, Intermediate Progress, and Final — due at different points in the semester. Each later report should include (a short, updated version of) the content from the earlier report(s), so that the Final Report alone tells the complete story of your problem, baseline, analysis, and improvement.

You are **free to choose any topic in computer vision except for the filtering contents** in assignment1. A list of example topics is provided below — these are topics the professor and MMLAB members are already familiar with, so you will get more specific and useful feedback if you pick one of them, but you are not required to. In addition, even if you choose one of the topics above, you are not required to use the listed reference paper.

Topic Example

- Zero-Shot Image Captioning
  - CapDec (EMNLP 2022 Findings https://arxiv.org/abs/2211.00575)
  - ViECap(ICCV 2023 https://arxiv.org/pdf/2307.16525)
  - SynTIC (ICCV2023 https://arxiv.org/pdf/2312.08865)
- Image Editing
  - SDEdit (ICLR2022 https://arxiv.org/abs/2108.01073)
  - Prompt-to-Prompt (ICLR 2023 https://arxiv.org/abs/2208.01626)
  - InstructPix2Pix (CVPR 2023 https://arxiv.org/abs/2211.09800)
- Video Retrieval
  - X-Pool (CVPR 2022 https://arxiv.org/pdf/2203.15086)
  - UCoFiA (ICCV2023 https://arxiv.org/abs/2309.10091)
  - Cap4Video (CVPR 2023 https://arxiv.org/abs/2301.00184)
- Video Captioning
  - VideoMAE (NeurIPS 2021 Spotlight https://arxiv.org/abs/2203.12602)
  - PDVC (ICCV 2021 https://arxiv.org/pdf/2108.07781)
  - CM^2(CVPR 2024 https://arxiv.org/pdf/2404.07610)
- Moment Retrieval and Highlight Detection
  - Moment-DETR (NeurIPS 2021 https://arxiv.org/abs/2107.09609)
  - UMT (CVPR 2022 https://arxiv.org/pdf/2203.12745)
  - QD-DETR (CVPR 2023 https://arxiv.org/pdf/2303.13874)

Criterion about Dataset

In this project, students are given considerable flexibility in choosing the topic and the scale of the dataset. However, to ensure fair evaluation, students are expected to **reproduce the baseline paper as faithfully as possible**. Therefore, all reports must clearly describe the selected benchmark dataset from the baseline paper, including its size, preprocessing procedures, and train/validation/test split criteria. **Using an arbitrary subset or sampling the data differently from the baseline without proper justification is not allowed**.

If sampling is necessary due to the dataset size or computational constraints, students **must clearly report the sampling criteria, the sampling ratio, and the justification**. Students must also report the computational environment and its actual usage — GPU model and memory, batch size, number of epochs, memory usage, and total training time — so that the reduced scale can be verified against the resources available. Even in this case, the selected data scale should make **sufficient use of the available computational resources**. The minimum expected computational environment is based on the free resources provided by Google Colab and Kaggle.

In summary, evaluation about dataset will focus on **whether students make sufficient use of their available data and computational resources to reproduce the baseline as faithfully as possible through the project**.

About Report

1. Proposal Report [Total: 5pt] : November 15th 11:59 PM

- **Topic & Dataset [2pt]** : what problem you are tackling, why it is interesting, and which dataset you will use (source, size, and any license/preprocessing notes). Be sure to follow the Dataset Criterion above when selecting the dataset scale and any sampling strategy. (If you don't write in detail, you don't get any score.)
- **Related paper review [2pt]** : summarize the baseline paper & related paper you found — what problem it addresses and how it solves it previously (its method, at a high level).
- **Plan [1pt]** : briefly describe how you will apply/reproduce this baseline on your chosen dataset (including any sampling/scaling-down plan).

2. Intermediate Progress Report [Total: 5pt] : **November 29th 11:59 PM**

- Include an updated, short version of the Proposal Report content.
- **Baseline reproduction [2pt]** : report the results you obtained by running the baseline on the selected benchmark dataset (quantitative numbers + qualitative examples). Clearly describe all details necessary to reproduce your experiment.
- **Limitation analysis & initial idea [3pt]**: based on your reproduction, discuss where and why the baseline fails or falls short (e.g., failure cases). Propose an approach to address one of the limitations and show preliminary results(a full solution is not required yet, showing the direction is enough). Clearly state whether the approach is your own idea or is adapted from existing work (both are allowed, but the source must be cited).

3. Final Report [Total: 10pt] : **December 18th 11:59 PM**

- Include an updated, short version of the Intermediate Progress Report content with high quality.
- Summarize topic and ideas you tried to improve on the baseline — including ones that did not work — and clearly identify which method performed best and why.
- Report your best-performing method and show that it **outperforms** the reproduced baseline under the same split and metrics (include quantitative and qualitative results). **Improvement is required; your own method receives additional credit.**
- Analyze the results using appropriate evidence, such as visualizations, quantitative analysis, and qualitative analysis.
- Discuss the limitations that still remain with your final method and suggest possible future directions